

Kubernetes Error 7: Service Not Accessible

What does "Service Not Accessible" mean?

- A Kubernetes Service is used to expose an application running inside pod. If the service is not accessible, users or other applications cannot reach the application.
- Common symptoms:
 -  Unable to access the application
 -  curl returns Connection Refused
 -  Browser shows 404, 503, or Timeout
 -  LoadBalancer EXTERNAL-IP is not responding

Kubernetes Service Not Accessible — Common Reasons & Solutions

 Cause	 Description	 Solution	 Useful Commands
 Selector Mismatch	Service selector does not match Pod labels	Verify the Service selector and Pod labels	<code>kubectl get svc</code> <code>kubectl get pods --show-labels</code>
 Incorrect targetPort	targetPort does not match the container's listening port	Update targetPort to match the application's port	<code>kubectl describe svc <service-name></code>
 Wrong Service Port	Service port is configured incorrectly	Verify the Service port and client connection port	<code>kubectl get svc</code>
 Pods Not Running / Not Ready	Backend Pods are not running or not Ready	Fix Pod issues and ensure readiness probes succeed	<code>kubectl get pods</code>
 Network Policy	NetworkPolicy blocks traffic to the Pods	Update NetworkPolicy to allow the required traffic	<code>kubectl get networkpolicy</code>
 Firewall / Security Group	External traffic to NodePort or	Open the required ports in the firewall	Check firewall or cloud security

 Cause	 Description	 Solution	 Useful Commands
	LoadBalancer is blocked	or cloud security group	group rules
 No Endpoints	Service has no backend Pods	Ensure Pods are running and labels match the Service selector	kubectl get endpoints <service-name>
 DNS Resolution Issue	Service name cannot be resolved	Verify CoreDNS is healthy and use the correct Service name	kubectl get pods -n kube-system
 LoadBalancer Pending	External IP has not been assigned	Verify the cloud controller manager and LoadBalancer provisioning	kubectl get svc

Service Not Accessible Troubleshooting Flow

Application Not Accessible

↓
kubectl get svc

↓
kubectl describe svc <service-name>

↓
Selector Matches Pod Labels?

↓
Yes

↓
Check Endpoints

↓
Endpoints Available?

↓
Yes

↓
Verify targetPort = containerPort

↓
Test Using BusyBox Pod

↓
Works Inside Cluster?

↓
Yes

↓
Check Firewall,
LoadBalancer,
Ingress

↓
No

↓
Check Pods,
Labels,
Application Logs

ApplicationAccessible



Useful Troubleshooting Commands

 Command	 Purpose
<code>kubectl get svc</code>	View Service details
<code>kubectl describe svc <service-name></code>	Inspect Service configuration
<code>kubectl get pods --show-labels</code>	Verify Pod labels
<code>kubectl get endpoints <service-name></code>	Check backend endpoints
<code>kubectl get pods</code>	Verify Pod status
<code>kubectl describe pod <pod-name></code>	Check Pod readiness and events
<code>kubectl get networkpolicy</code>	List NetworkPolicies
<code>kubectl exec -it <pod-name> -- curl <service-name>:<port></code>	Test Service connectivity from inside the cluster
<code>kubectl get events --sort-by=.lastTimestamp</code>	Review recent cluster events
<code>kubectl logs <pod-name></code>	Check application logs
<code>kubectl get networkpolicy</code>	Check for network restrictions

Interview Answer

Q: How do you troubleshoot a Kubernetes Service that is not accessible?

- Check the Service configuration using `kubectl describe svc`.
- Verify that the Service selector matches the Pod labels.
- Ensure the backend Pods are running and in the Ready state.
- Confirm the Service has endpoints using `kubectl get endpoints`.
- Verify the `port` and `targetPort` values.
- Check for NetworkPolicy restrictions.
- For NodePort or LoadBalancer Services, verify `firewall` or `cloud security group` rules.

Simple Answer

-  A Kubernetes Service is usually inaccessible because of **selector mismatches**, **incorrect target Port** or **port configuration** , **unhealthy backend Pods** , **missing endpoints**, **Network Policy restrictions** , or **blocked NodePort** / **LoadBalancer traffic**
-  The first step is to inspect the Service with `kubectl describe svc` and verify its endpoints using `kubectl get endpoints`.